

Macro I

NAME:

Problem Set 4

Submit Q1-Q2; TA session will cover Q3-Q4; Q5 is for extra practice

Question 1: Workforce size and optimal investment

Assume that firms make investment decisions according to the model we discussed in class. Specifically, think about a two-period model and the following assumptions:

- $Y_t = A_t(K_t)^{0.35}(N_t)^{0.65}$ (note: this is the production function for every period)
- $A = A^f = 5$
- $N = 30$ and $N^f = 30$
- $p_k = 5$ and $p_k^f = 5.1$
- $r = 0.03$
- $\delta = 0.1$
- $\tau_k = 0.2$
- $K = 100$

1. Find the optimal level of K^f and the optimal level of current I . For this, follow the usual steps:
 - (a) What is the value of the user cost?
 - (b) Derive a term for MPK^f
 - (c) Equate the two to have one equation with one unknown K^f , and solve it.
 - (d) Use the capital accumulation equation to solve for I
2. Assume that the goods market is in equilibrium. Calculate the levels of saving and consumption in the economy in the current period.
3. Assume that the labor input rises temporarily (say due to temporary work visas) so that $N = 40$ but $N^f = 30$.
 - (a) Calculate the levels of GDP and GDP per worker before and after the change. Do GDP and GDP per worker increase or decrease? Briefly explain why.
 - (b) Is there a new level of optimal capital in the future (K^f)? If so, calculate it. If not, explain why not.
 - (c) What are the levels of GDP and GDP per worker in the future period?

4. Now assume that the labor input rises permanently (say due to immigration or increased labor force participation) so that $N = 40$ and $N^f = 40$.
 - (a) Calculate GDP in the current period. How does it compare to the GDP you calculated in part 2 and part 3(a)?
 - (b) Is there a new level of optimal capital in the future (K^f)? If so, calculate it. If not, explain why not.
 - (c) What is the level of GDP in the future period?
5. Briefly explain the difference in results between parts (3) and (4).

Question 2 - Optimism and Goods Market Equilibrium

Some economists claim that economic fluctuations can be caused by “waves of optimism” (or pessimism) about the future prospects of the economy. In this question we analyze one example that illustrates such a claim using our goods market equilibrium model.

Suppose that an economy is at a current state of equilibrium where consumers solve their optimal plan, firms know their optimal K^f and investment. In equilibrium, $S = I$ and there is some market clearing real interest rate r .

Suppose that producers (firms) in this economy suddenly become optimistic about the future prospects of productivity, i.e. they believe that A^f will be higher than what was initially expected. For simplicity, also assume that consumers *do not* share this sentiment, and nothing changes in their beliefs.

Briefly answer the questions below, and provide graphs and explanations (where applicable).

1. What is the implication, if any, with regards to savings? Does the S curve shift to the left, to the right, or does not shift?
2. What is the implication, if any, with regards to investment? Does the I curve shift to the left, to the right, or does not shift?
3. As a result of your previous answers, describe the new equilibrium in the goods market: what happens to I, S, r at the new equilibrium?
4. Going back to the claim about expectations and fluctuations, explain how the change in future output is consistent with the claim. Explain how the change in current consumption is inconsistent with the claim.

Question 3 - for TA session - taxes and their use

Assume that firms make investment decisions according to the model we discussed in class. Specifically, think about a two-period model and the following assumptions:

- $Y_t = A_t(K_t)^{0.3}(N_t)^{0.7}$
- $A = A^f = 125$
- $N = 500$ and $N^f = 500$

- $p_k = 120$ and $p_k^f = 100$
 - $r = 0.04$
 - $\delta = 0.1$
 - $\tau_k = 0.1$
 - $K = 400$
1. Find the optimal level of K^f and the optimal level of current I . For this, follow the usual steps:
 - (a) What is the value of the user cost?
 - (b) Derive a term for MPK^f
 - (c) Equate the two to have one equation with one unknown K^f , and solve it.
 - (d) Use the capital accumulation equation to solve for I
 2. Assume that the government decides to increase the tax rate τ_k from 0.1 to 0.15. Assume that the rest of the parameters remain unchanged. Find the optimal level of K^f and the optimal level of current I . (For this, follow the same steps as before).
 3. Now assume that the government uses the extra tax proceeds to improve infrastructure that will increase future productivity, i.e. A^f increases to 150. Find the optimal level of K^f and the optimal level of current I . (again, follow the same steps as before).
 4. Briefly explain the difference in results between parts (2) and (3).

Question 4 - For TA Session - Effect of Government Spending on Goods Market Equilibrium

Suppose that an economy is at a current state of equilibrium where consumers solve their optimal plan, firms know their optimal K^f and investment. In equilibrium, $S = I$ and there is some market clearing real interest rate r .

Suppose that the government announces (unexpectedly) that it is going to increase the current level of government expenditure (G). In order to finance the additional spending, the government also announces a one time (or one period) tax increase of the same amount. For simplicity, assume that the tax is imposed “lump-sum”, i.e. a fixed amount that will be paid by all consumers in the economy. You may also assume that consumers do not enjoy G in any way, and that the government spending is not repaid to the consumers in any way.

Briefly answer the questions below, and provide graphs and explanations (where applicable).

1. What does the shock imply with regards to consumers' *current* disposable income? consumers' PVLR?
2. If consumers behave according to our standard consumption model, how should their consumption respond? will it increase/decrease? by more/less/same as the change in government spending.

3. What is the implication, if any, with regards to savings? Does the S curve shift to the left, to the right, or does not shift?
4. What is the implication, if any, with regards to investment? Does the I curve shift to the left, to the right, or does not shift?
5. As a result of your previous answers, describe the new equilibrium in the goods market: what happens to I, S, r at the new equilibrium?

Question 5 - Extra

Assume that firms make investment decisions according to the model we discussed in class, and that the production function is a standard Cobb-Douglas $Y = AK^\alpha N^{1-\alpha}$ (where $0 < \alpha < 1$ is a parameter, $A > 0$ is total factor productivity, $K > 0$ is capital, and $N > 0$ is the labor input). Also assume that the user cost is some constant number X .

1. Show that if firms expect to have higher labor input in the future period, then they choose to have more capital for the future period.
2. Will the same result hold if the production function is $Y = A[K^\alpha + N^{1-\alpha}]$? prove and explain your claim.

Solution

1. Let's follow the same steps as in class. We know (by assumption) that user cost = X . The marginal product of capital in the future period is:

$$MPK^f = \frac{\partial Y^f}{\partial K^f} = \frac{\partial A^f (K^f)^\alpha (N^f)^{1-\alpha}}{\partial K^f} = \alpha A^f (K^f)^{\alpha-1} (N^f)^{1-\alpha}$$

Now set $MPK^f = X$ and solve for K^f as a function of all the other variables:

$$\begin{aligned} \alpha A^f (K^f)^{\alpha-1} (N^f)^{1-\alpha} &= X \\ (K^f)^{\alpha-1} &= \frac{X}{\alpha A^f (N^f)^{1-\alpha}} \\ (K^f)^{1-\alpha} &= \frac{\alpha A^f (N^f)^{1-\alpha}}{X} \\ K^f &= \left[\frac{\alpha A^f}{X} \right]^{\frac{1}{1-\alpha}} N^f \end{aligned}$$

Since the term in brackets is positive then it is clear that $\frac{\partial K^f}{\partial N^f} > 0$, which is what we need for proving the claim in the question.

2. In this case the future marginal product of capital is:

$$MPK^f = \frac{\partial Y^f}{\partial K^f} = \frac{\partial A^f [(K^f)^\alpha + (N^f)^{1-\alpha}]}{\partial K^f} = \alpha A^f (K^f)^{\alpha-1}$$

and therefore:

$$\begin{aligned}
\alpha A^f (K^f)^{\alpha-1} &= X \\
(K^f)^{\alpha-1} &= \frac{X}{\alpha A^f} \\
(K^f)^{1-\alpha} &= \frac{\alpha A^f}{X} \\
K^f &= \left[\frac{\alpha A^f}{X} \right]^{\frac{1}{1-\alpha}}
\end{aligned}$$

And now we have a case where K^f is independent of N^f and we can conclude that N^f does not affect the optimal capital decision.

The reason for this results is that the alternative production function that we consider does not satisfy the complementarity assumption that we typically require and the that the Cobb-Douglas function satisfies.